# Executive Summary

## Problem:

- PowerCo is experiencing customer churn, and they suspect it might be due to customers being sensitive to prices. To address this, they are considering offering a 20% discount to customers who are at high risk of leaving as a potential solution.

## Machine Learning Modelling:

- Following data cleaning, exploratory data analysis, and feature engineering, I used a Random Forest Classifier to predict the likelihood of customers churning. The baseline model achieved a test set accuracy of 0.90 and a precision score of 0.88, but caught only 6% of churners (recall 0.06) because so few customers churn.

- Re-training with class weighting and a decision threshold chosen on training data raised recall to 0.28, about 4.6x more at-risk customers identified.

## Insights:

- Churn rate is approximately 9.7%
- Price sensitivity alone has only a weak correlation with churn
- Net margin on power subscription and consumption over the last 12 months are significant factors for churn
- The forecasted bill for meter rental is also an influential factor for churn
- Time-related factors such as the number of months of activity also influence churn

## Recommendation:

- Offering the 20% discount to customers with a predicted churn probability of 0.5 or above gave the largest estimated revenue gain on the test set (about $2,930). Because price is a weak driver, the discount should be combined with other retention efforts, and collecting data on competitor prices and customer service interactions would improve the model.